

Project 9: Blind Blur Estimation for Image Restoration

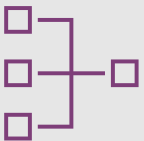
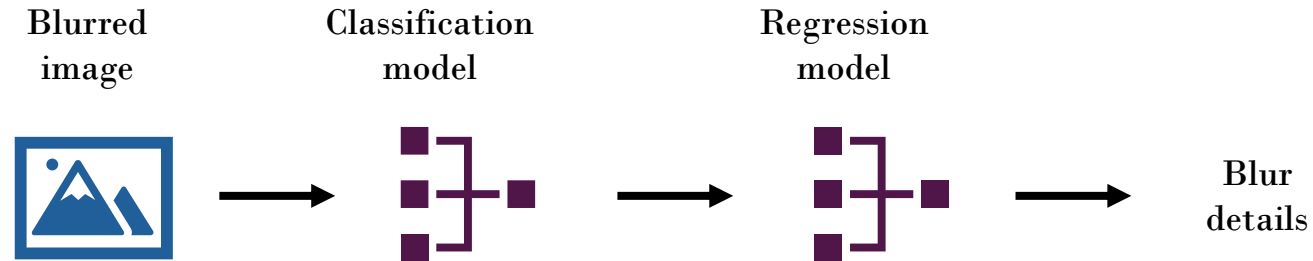
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Introduction

< Blind Blur Estimation for Image Restoration >

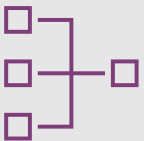
The aim of the project is to build a pipeline of two AI models for the purpose of recognizing the blur affecting an input image. The work is based on the paper "***Blind Image Blur Estimation via Deep Learning***," by R. Yan and L. Shao.



Project Task

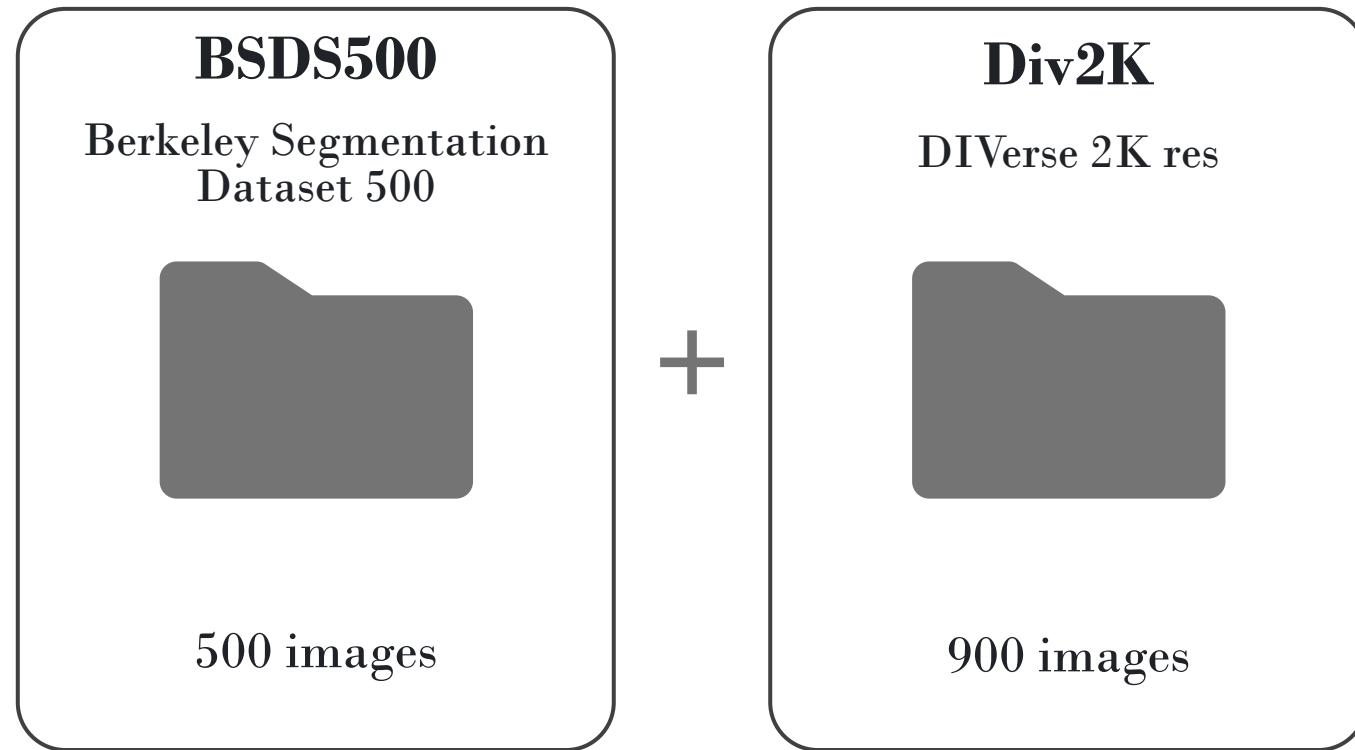
1. Dataset generation (image cropping and blurring)
2. Implementation of two models  

CNN + GRNN
MobileViT + GRNN
3. Performance comparison and evaluation between the models

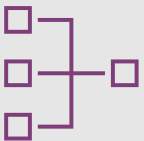


Dataset

Available data for our project:

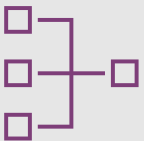
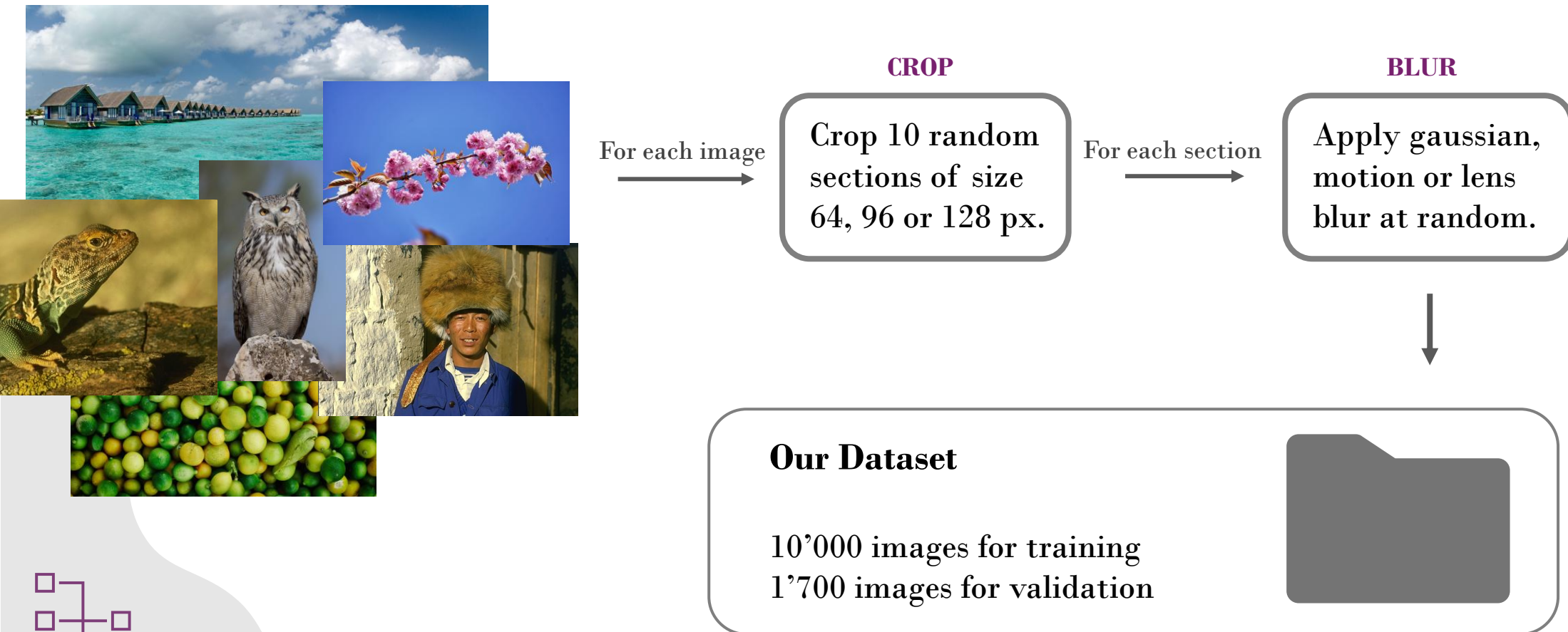


Overall a pretty small training dataset.



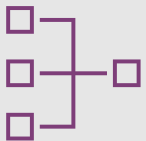
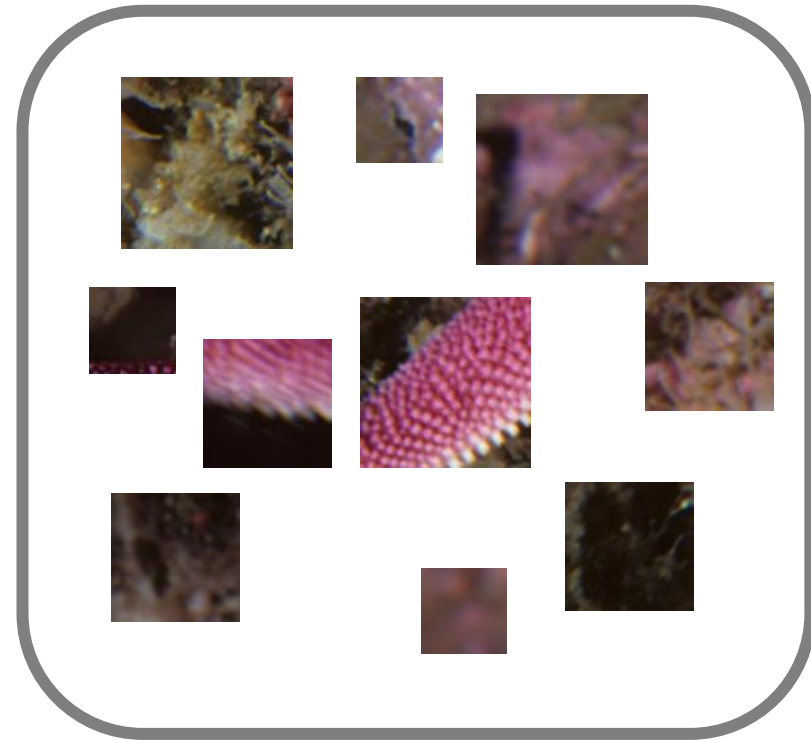
Dataset

We use **data augmentation** to create a bigger dataset to train our models on:



Dataset

Here is an example of how the different images are being cropped and blurred



Models overview – Resnet18

Convolutional neural network

The resnet models are pretty simple CNNs, perfect for image classification and object detection because of its main features:

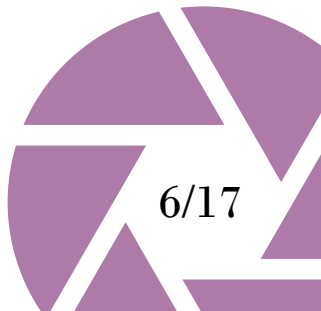
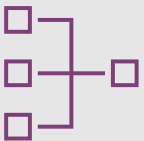
- **Easy to optimize**
- **Easy to fine-tune**, thanks to pre-trained parameters
- **Lightweight**



Training data augmentation

40% of the data is augmented with:

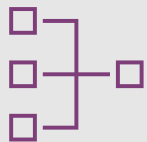
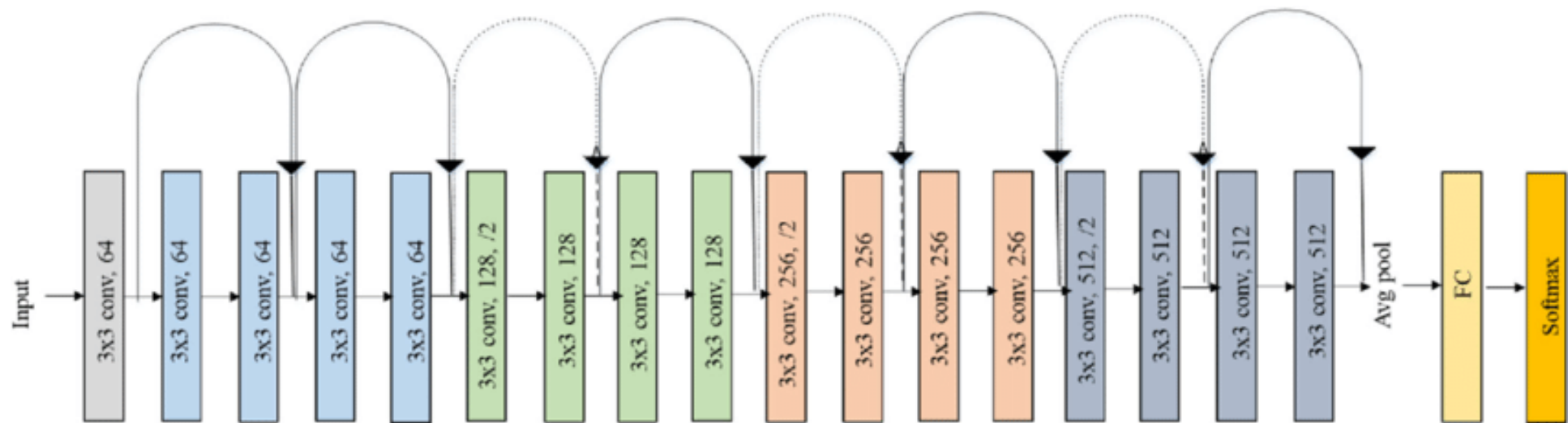
- Random masking
- Gaussian noise
- Color jittering



Resnet18 - Architecture

Convolutional neural network

The resnet architecture remains the same, apart for the 3-node output layer.

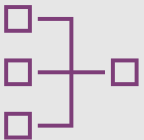


Models overview – GRNN

General regression neural network

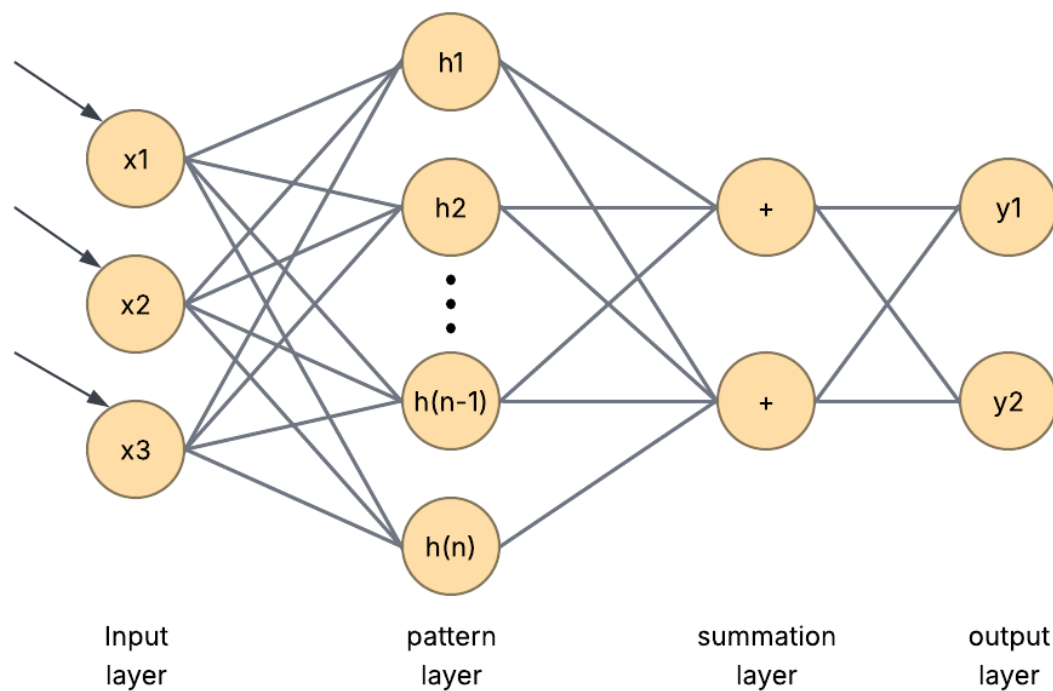
Out of the ordinary network that uses a gaussian distance function to compute the most probable output based on the classification input. Main features includes:

- **No iterative training**
- **High noise robustness**
- **Network size** depends on training pool size



GRNN - Architecture

General regression neural network



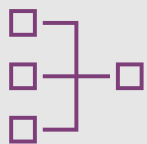
Blur parameters in output:

Kernel size (of the blur applied to the image).

Size can vary between: 3, 5, 7 and 9.

Output value depends on the blur:

Gaussian	Standard deviation
Motion	Motion angle
Lens	-



Models overview – MobileViT

Transformer encoder

MobileViT is a **lightweight Visual Transformer** architecture defined by:

- **Compact and efficient design**, mostly used for mobile application,
- **Hybrid structure**, with two different types of block in order to mix convolution and global transformer operations.
- **CNN spatial biases**, used for efficiency.

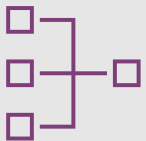


**Training data
augmentation +
frequency conversion**

40% of the data is augmented with:

- Random masking
- Gaussian noise
- Color jittering

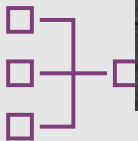
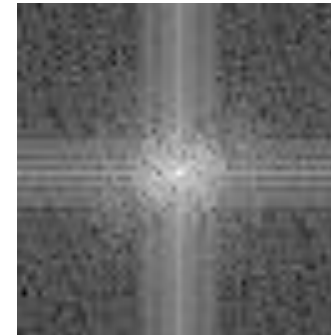
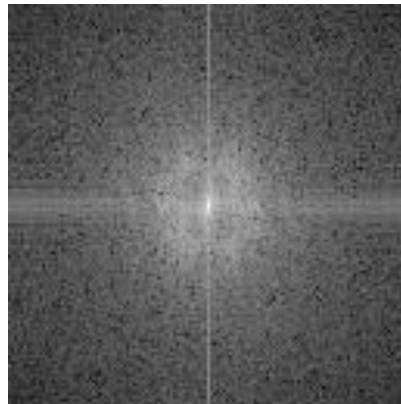
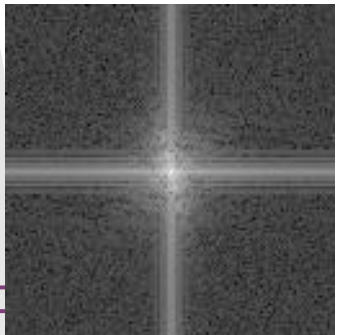
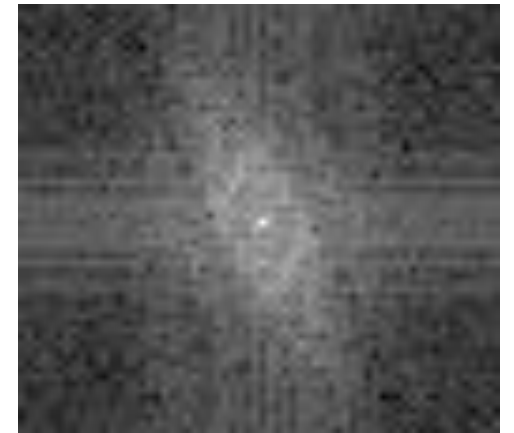
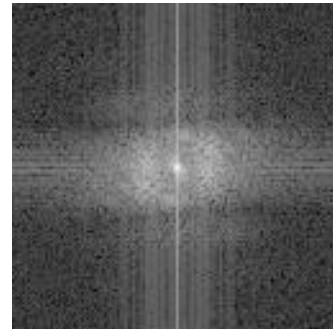
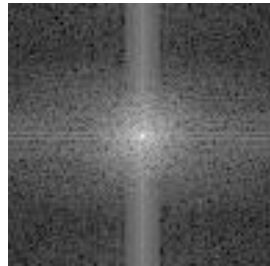
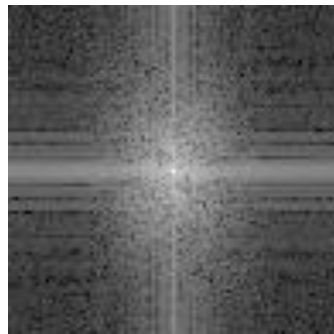
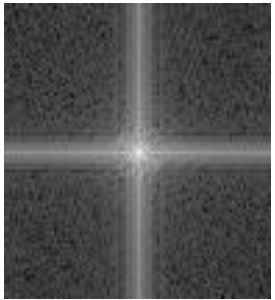
Furthermore, images are converted into frequency domain before being passed to the model.



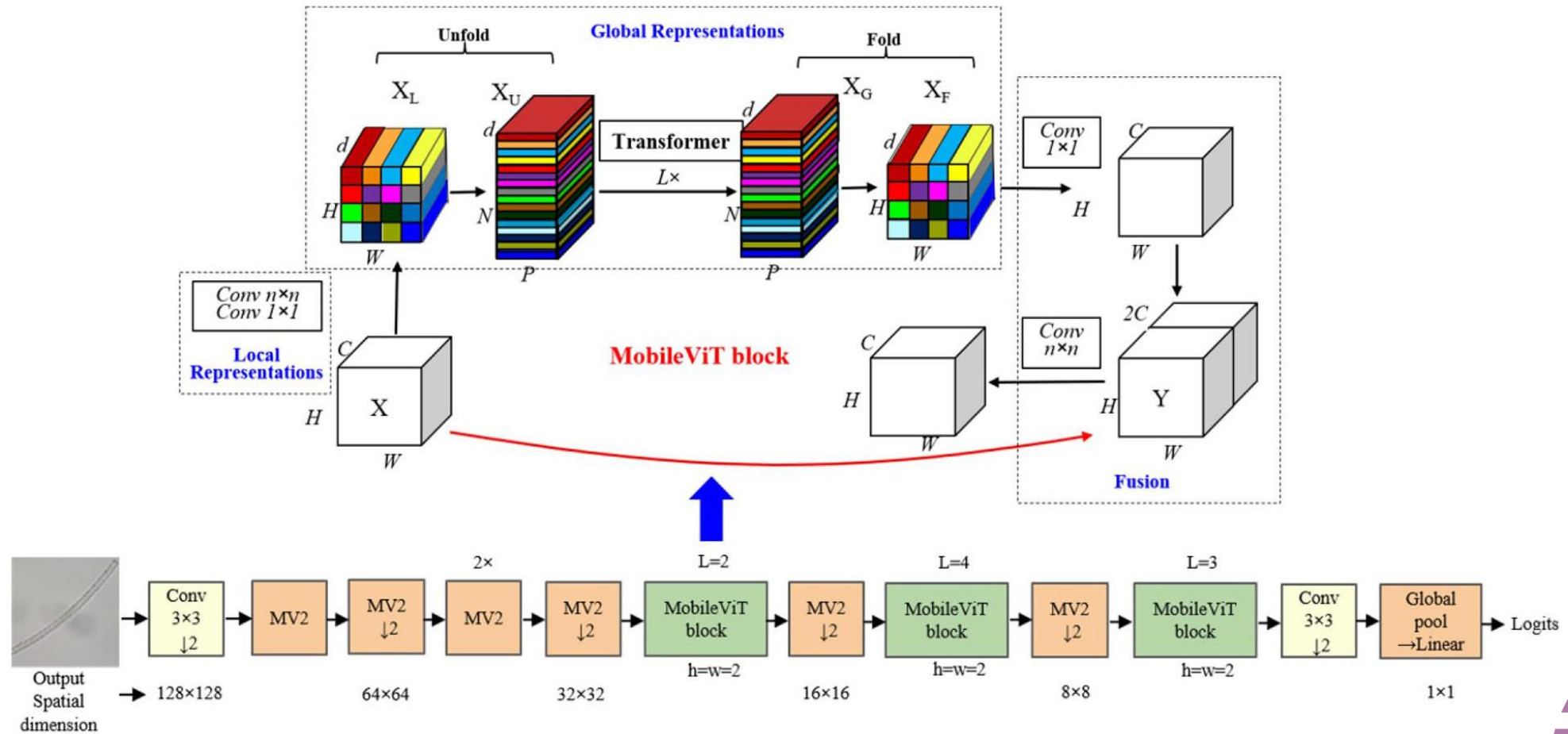
Models overview – MobileVit

Transformer encoder

This is an example of the frequency conversion of some of the images on slide 5:



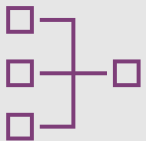
Models overview – MobileViT



Testing environment

All the training and testing have been done with desktop pc, with windows 11, intel i9 14900k CPU, 32GB ram ddr5 6000MHz and nVidia RTX5070Ti GPU.

	Training time	Epoch	Batch	Learning rate
Resnet	19 min	15	16	0.001
MobileVit	26 min	13	35	0.0005

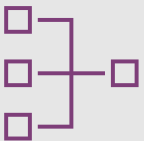


Results

Classification accuracy

	Gaussian	Motion	Lens	Total
Resnet	64%	89%	63%	70%
MobileVit	57%	90%	65%	67%

The models seems to have really similar results in classification.



Results

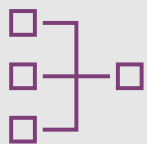
Confusion matrix, made with the evaluation pool of about 1'700 samples. (mobile vit)

Predicted class

Real class

	Gaussian	Motion	Lens
Gaussian	541	31	63
Motion	231	350	95
Lens	161	7	290

The main diagonal resembles the true positives. The rest of the table represents the false positive of the predictions.



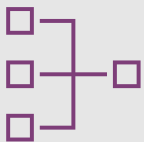
Results

Regression loss

	Kernel accuracy			Parameter accuracy		Total
	Gaussian	Motion	Defocus	Gaussian (sigma)	Motion (angle)	
GRNN RES	0.45	0.53	0.45	0.28	0.37	0.44
GRNN ViT	0.17	0.22	0.187	0.26	0.3	0.25

Loss computed on normalized values between 0 and 1. Every value is the root of the MSE loss function.

We can see the GRNN associated with the ViT transformer performs better.



End

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